

$$\begin{aligned}
 &= L^{-1} \left\{ \frac{4s}{(s^2 + 2)^2 - (2s)^2} \right\} = L^{-1} \left\{ \frac{4s}{(s^2 + 2 + 2s)(s^2 + 2 - 2s)} \right\} \\
 &= L^{-1} \left\{ \frac{1}{s^2 - 2s + 2} - \frac{1}{s^2 + 2s + 2} \right\} = L^{-1} \left\{ \frac{1}{(s-1)^2 + 1} - \frac{1}{(s+1)^2 + 1} \right\} \\
 &= e^t \sin t - e^{-t} \sin t = 2 \sinh t \sin t.
 \end{aligned}$$

21.14 CONVOLUTION THEOREM

If $L^{-1} \{ \bar{f}(s) \} = f(t)$, and $L^{-1} \{ \bar{g}(s) \} = g(t)$,

then $L^{-1} \{ \bar{f}(s) \bar{g}(s) \} = \int_0^t f(u) g(t-u) du = F * G$

[$F * G$ is called the **convolution** or **fasting** of F and G .]

Let $\phi(t) = \int_0^t f(u) g(t-u) du$

$$L\{\phi(t)\} = \int_0^\infty e^{-st} \left\{ \int_0^t f(u) g(t-u) du \right\} dt = \int_0^\infty \int_0^t e^{-st} f(u) g(t-u) du dt \quad \dots(1)$$

The domain of integration for this double integral is the entire area lying between the lines $u = 0$ and $u = t$ (Fig. 21.2).

On changing the order of integration, we get

$$\begin{aligned}
 L\{\phi(t)\} &= \int_0^\infty \int_u^\infty e^{-st} f(u) g(t-u) dt du \\
 &= \int_0^\infty e^{-su} f(u) \left\{ \int_0^\infty e^{-s(t-u)} g(t-u) dt \right\} du \\
 &= \int_0^\infty e^{-su} f(u) \left\{ \int_0^\infty e^{-sv} g(v) dv \right\} du \text{ on putting } t-u=v \\
 &= \int_0^\infty e^{-su} f(u) g(s) du = \int_0^\infty e^{-su} f(u) du \cdot \bar{g}(s) \\
 &= \bar{f}(s) \cdot \bar{g}(s) \text{ whence follows the desired result.}
 \end{aligned}$$

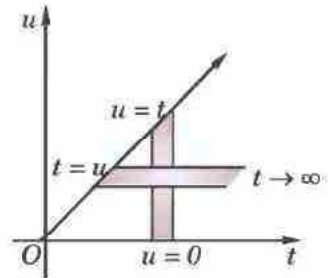


Fig. 21.2

Example 21.25. Apply Convolution theorem to evaluate

(i) $L^{-1} \frac{s}{(s^2 + a^2)^2}$

(V.T.U., 2010)

(ii) $L^{-1} \frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$

(V.T.U., 2011 S ; Bhopal, 2008 ; Mumbai, 2007)

Solution. (i) Since $f(t) = L^{-1} \left(\frac{s}{s^2 + a^2} \right) = \cos at$ and $g(t) = L^{-1} \left(\frac{s}{s^2 + a^2} \right) = \frac{1}{a} \sin at$

$\therefore$ by Convolution theorem, we get

$$\begin{aligned}
 L^{-1} \left[\frac{s}{s^2 + a^2} \cdot \frac{1}{s^2 + a^2} \right] &= \int_0^t \cos au \frac{\sin a(t-u)}{a} du \\
 &= \frac{1}{2a} \int_0^t [\sin at - \sin (2au - at)] dt = \frac{1}{2a} \left[u \sin at + \frac{1}{2a} \cos (2au - at) \right]_0^t = \frac{1}{2a} t \sin at
 \end{aligned}$$

Hence $L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\} = \frac{1}{2a} t \sin at.$

(ii) Since $f(t) = L^{-1} \left(\frac{s}{s^2 + a^2} \right) = \cos at$ and $g(t) = L^{-1} \left(\frac{s}{s^2 + b^2} \right) = \cos bt$,

$\therefore$ by Convolution theorem, we get

$$\begin{aligned} L^{-1} \left(\frac{s}{s^2 + a^2} \cdot \frac{s}{s^2 + b^2} \right) &= \int_0^t \cos au \cos b(t-u) du \quad [\because f(u) = \cos au, g(t-u) = \cos b(t-u)] \\ &= \frac{1}{2} \int_0^t [\cos [(a-b)u + bt] + \cos [(a+b)u - bt]] du \\ &= \frac{1}{2} \left[\frac{\sin [(a-b)u + bt]}{a-b} + \frac{\sin [(a+b)u - bt]}{a+b} \right]_0^t \\ &= \frac{1}{2} \left\{ \frac{\sin at - \sin bt}{a-b} + \frac{\sin at + \sin bt}{a+b} \right\} = \frac{a \sin at - b \sin bt}{a^2 - b^2}. \end{aligned}$$

Example 21.26. Evaluate (i) $L^{-1} \frac{1}{(s^2 + 1)(s^2 + 9)}$

(Mumbai, 2005 S)

(ii) $L^{-1} \frac{s}{(s^2 + 1)(s^2 + 4)(s^2 + 9)}$.

(Madras, 2006)

Solution. (i) Since $L^{-1} \left(\frac{1}{s^2 + 1} \right) = \sin t$, $L^{-1} \left(\frac{1}{s^2 + 9} \right) = \frac{\sin 3t}{3}$

$\therefore$ by Convolution theorem, we get

$$\begin{aligned} L^{-1} \left(\frac{1}{s^2 + 1} \cdot \frac{1}{s^2 + 9} \right) &= \int_0^t \sin u \cdot \frac{\sin 3(t-u)}{3} du \\ &= \frac{1}{6} \int_0^t [\cos (4u - 3t) - \cos (3t - 2u)] du = \frac{1}{6} \left[\frac{\sin (4u - 3t)}{4} - \frac{\sin (3t - 2u)}{-2} \right]_0^t \\ &= \frac{1}{6} \left\{ \frac{1}{4} (\sin t + \sin 3t) + \frac{1}{2} (\sin t - \sin 3t) \right\} = \frac{1}{8} (\sin t - \frac{1}{3} \sin 3t) \end{aligned}$$

(ii) Since $L^{-1} \left(\frac{s}{s^2 + 4} \right) = \cos 2t$ and $L^{-1} \left\{ \frac{1}{(s^2 + 1)(s^2 + 9)} \right\} = \frac{1}{8} \left[\sin t - \frac{1}{3} \sin 3t \right]$

[By (i)]

$\therefore$ by Convolution theorem, we get

$$\begin{aligned} L^{-1} \frac{s}{(s^2 + 1)(s^2 + 4)(s^2 + 9)} &= L^{-1} \left\{ \frac{1}{(s^2 + 1)(s^2 + 9)} \cdot \frac{s}{s^2 + 4} \right\} \\ &= \int_0^t \frac{1}{8} (\sin u - \frac{1}{3} \sin 3u) \cdot \cos 2(t-u) du \\ &= \frac{1}{8} \int_0^t [\sin u \cos 2(t-u) - \frac{1}{3} \sin 3u \cos 2(t-u)] du \\ &= \frac{1}{8} \int_0^t \left[\frac{1}{2} \{\sin (2t-u) - \sin (3u-2t)\} - \frac{1}{6} \{\sin (u+2t) - \sin (5u-2t)\} \right] du \\ &= \frac{1}{16} \left[\left. \frac{-\cos (2t-u)}{-1} + \frac{\cos (3u-2t)}{3} \right|_0^t \right] - \frac{1}{48} \left[\left. -\cos (u+2t) + \frac{\cos (5u-2t)}{5} \right|_0^t \right] \\ &= \frac{1}{12} \cos t - \frac{1}{10} \cos 2t + \frac{1}{60} \cos 3t. \end{aligned}$$

PROBLEMS 21.5

Find the inverse transforms of :

1. $\frac{1}{s^2(s+5)}$. (Madras, 2003 S)
2. $\frac{1}{s(s+2)^3}$.
3. $\frac{s}{a^2s^2+b^2}$. (Madras, 2000 S)
4. $\frac{1}{s^2(s^2+a^2)}$.
5. $\frac{1}{s^3(s^2+1)}$.
6. $\frac{s+2}{(s^2+4s+8)^2}$. (Mumbai, 2006)
7. $\frac{2as}{(s^2+a^2)^2}$.
8. $\frac{s^2}{(s+a)^3}$.
9. $\log\left(\frac{1+s}{s}\right)$.
10. $\log\left(\frac{s+a}{s+b}\right)$. (Anna, 2003 ; U.P.T.U., 2003)
11. $\log\left\{\frac{s+1}{(s+2)(s+3)}\right\}$.
12. $\frac{1}{2}\log\left(\frac{s^2+b^2}{s^2+a^2}\right)$. (Mumbai, 2008 ; V.T.U., 2008)
13. $\log\left(1-\frac{a^2}{s^2}\right)$.
14. $\log\frac{s^2+1}{(s-1)^2}$. (Madras, 2000 S)
15. $\tan^{-1}\left(\frac{2}{s}\right)$ (Mumbai, 2007 ; P.T.U., 2005)
16. $\cot^{-1}(s)$. (V.T.U., 2005)
17. $s \log \frac{s-1}{s+1}$ (Madras, 1999)

Using Convolution theorem, evaluate :

18. $L^{-1}\left\{\frac{1}{(s+a)(s+b)}\right\}$.
19. $L^{-1}\frac{1}{(s^2+a^2)^2}$.
20. $L^{-1}\frac{1}{s^2(s^2+a^2)}$.
21. $L^{-1}\left\{\frac{1}{s^2(s+1)^2}\right\}$.
22. $L^{-1}\left\{\frac{1}{(s-2)(s+2)^2}\right\}$. (Mumbai, 2009)
23. $L^{-1}\left\{\frac{s}{(s+2)(s^2+9)}\right\}$. (V.T.U., 2008 S)
24. $\frac{1}{s^3(s^2+1)}$. (V.T.U., 2007 ; U.P.T.U., 2005)
25. $\frac{1}{(s^2+4s+13)^2}$. (Mumbai, 2008)
26. Show that (i) $L^{-1}\left(\frac{1}{s}\sin\frac{1}{s}\right) = t - \frac{t^3}{(3!)^2} + \frac{t^5}{(5!)^2} - \frac{t^7}{(7!)^2} + \dots$
 (ii) $L^{-1}\left(\frac{1}{s}\cos\frac{1}{s}\right) = 1 - \frac{t^2}{(2!)^2} + \frac{t^4}{(4!)^2} - \frac{t^6}{(6!)^2} + \dots$

21.15 (1) APPLICATION TO DIFFERENTIAL EQUATIONS

The Laplace transform method of solving differential equations yields particular solutions without the necessity of first finding the general solution and then evaluating the arbitrary constants. This method is, in general, shorter than our earlier methods and is specially useful for solving linear differential equations with constant coefficients.

(2) Working procedure to solve a linear differential equation with constant coefficients by transform method :

1. Take the Laplace transform of both sides of the differential equation using the formula of § 21.7, and the given initial conditions.

2. Transpose the terms with minus signs to the right.

3. Divide by the coefficient of $\bar{y}$, getting $\bar{y}$ as a known function of s .

4. Resolve this function of s into partial fractions and take the inverse transform of both sides. This gives y as a function of t which is the desired solution satisfying the given conditions.

Example 21.27. Solve by the method of transforms, the equation

$$y''' + 2y'' - y' - 2y = 0 \text{ given } y(0) = y'(0) = 0 \text{ and } y''(0) = 6.$$

(V.T.U., 2011 S)

Solution. Taking the Laplace transform of both sides, we get

$$[s^3 \bar{y} - s^2 y(0) - sy'(0) - y''(0)] + 2[s^2 \bar{y} - sy(0) - y'(0)] - [s \bar{y} - y(0)] - 2\bar{y} = 0$$

Using the given conditions, it reduces to

$$(s^3 + 2s^2 - s - 2) \bar{y} = 6$$

$$\therefore \bar{y} = \frac{6}{(s-1)(s+1)(s+2)} = \frac{6}{(s-1)(6)} + \frac{6}{(-2)(s+1)} + \frac{6}{3(s+2)}$$

On inversion, we get $y = L^{-1} \frac{1}{(s-1)} - 3L^{-1} \frac{1}{(s+2)} + 2L^{-1} \left(\frac{1}{s+2} \right)$

or $y = e^t - 3e^{-t} + 2e^{-2t}$ which is the desired result.

Example 21.28. Use transform method to solve

$$\frac{d^2 x}{dt^2} - 2 \frac{dx}{dt} + x = e^t \text{ with } x = 2, \frac{dx}{dt} = -1 \text{ at } t = 0.$$

(Anna, 2005 S)

Solution. Taking the Laplace transforms of both sides, we get

$$[s^2 \bar{x} - sx(0) - x'(0)] - 2[s \bar{x} - x(0)] + \bar{x} = \frac{1}{s-1}$$

Using the given conditions, it reduces to

$$(s^2 - 2s + 1) \bar{x} = \frac{1}{s-1} + 2s - 5 = \frac{2s^2 - 7s + 6}{s-1}$$

$$\therefore \bar{x} = \frac{2s^2 - 7s + 6}{(s-1)^3} = \frac{2}{s-1} - \frac{3}{(s-1)^2} + \frac{1}{(s-1)^3} \text{ on breaking into partial fractions.}$$

On inversion, we obtain $x = 2L^{-1} \left(\frac{1}{s-1} \right) - 3L^{-1} \frac{1}{(s-1)^2} + L^{-1} \frac{1}{(s-1)^3}$

$$= 2e^t - \frac{3e^t \cdot t}{1!} + \frac{e^t \cdot t^2}{2!} = 2e^t - 3te^t + \frac{1}{2}t^2e^t.$$

Example 21.29. Solve $(D^2 + n^2)x = a \sin(nt + \alpha)$, $x = Dx = 0$ at $t = 0$.

Solution. Taking the Laplace transforms of both sides, we get

$$[s^2 \bar{x} - sx(0) - x'(0)] + n^2 \bar{x} = aL[\sin nt \cdot \cos \alpha + \cos nt \cdot \sin \alpha]$$

On using the given conditions,

$$(s^2 + n^2) \bar{x} = a \cos \alpha \cdot \frac{n}{s^2 + n^2} + a \sin \alpha \cdot \frac{s}{s^2 + n^2}$$

$$\therefore \bar{x} = an \cos \alpha \cdot \frac{1}{(s^2 + n^2)^2} + a \sin \alpha \cdot \frac{s}{(s^2 + n^2)^2}$$

On inversion, we obtain

$$x = an \cos \alpha \cdot \frac{1}{2n^3} (\sin nt - nt \cos nt) + a \sin \alpha \cdot \frac{t}{2n} \sin nt \quad [\text{By (11) and (12) p. 741}]$$

$$= a \{ \sin nt \cos \alpha - nt \cos (nt + \alpha) \} / 2n^2.$$

Example 21.30. Solve $(D^3 - 3D^2 + 3D - 1)y = t^2 e^t$ given that $y(0) = 1$, $y'(0) = 0$, $y''(0) = -2$.

(S.V.T.U., 2009)

Solution. Taking the Laplace transforms of both sides, we get

$$[s^3 \bar{y} - s^2 y(0) - sy'(0) - y''(0)] - 3[s^2 \bar{y} - sy(0) - y'(0)] + 3[s \bar{y} - y(0)] - \bar{y} = \frac{2}{(s-1)^3}$$

Using the given conditions, it reduces to

$$\begin{aligned} \bar{y} &= \frac{s^2 - 3s + 1}{(s-1)^3} + \frac{2}{(s-1)^6} = \frac{(s-1)^2 - (s-1) - 1}{(s-1)^3} + \frac{2}{(s-1)^6} \\ &= \frac{1}{s-1} - \frac{1}{(s-1)^2} - \frac{1}{(s-1)^3} + \frac{2}{(s-1)^6} \end{aligned}$$

On inversion, we obtain $y = L^{-1} \left(\frac{1}{s-1} \right) - L^{-1} \frac{1}{(s-1)^2} - L^{-1} \frac{1}{(s-1)^3} + 2L^{-1} \frac{1}{(s-1)^6}$

$$= e^t \left(1 - t - \frac{1}{2}t^2 + \frac{1}{60}t^5 \right).$$

Example 21.31. Solve $\frac{d^2 x}{dt^2} + 9x = \cos 2t$, if $x(0) = 1$, $x(\pi/2) = -1$. (Bhopal, 2008; U.P.T.U., 2006)

Solution. Since $x'(0)$ is not given, we assume $x'(0) = a$.

Taking the Laplace transforms of both sides of the equation, we have

$$L(x'') + 9L(x) = L(\cos 2t) \text{ i.e., } [s^2 \bar{x} - sx(0) - x'(0)] + 9\bar{x} = \frac{s}{s^2 + 4}$$

$$(s^2 + 9)\bar{x} = s + a + \frac{s}{s^2 + 4} \quad \text{or} \quad \bar{x} = \frac{s+a}{s^2+9} + \frac{s}{(s^2+4)(s^2+9)}$$

or
$$\bar{x} = \frac{a}{s^2+9} + \frac{1}{5} \cdot \frac{s}{s^2+4} + \frac{4}{5} \cdot \frac{s}{s^2+9}.$$

On inversion, we get $x = \frac{a}{3} \sin 3t + \frac{1}{5} \cos 2t + \frac{4}{5} \cos 3t$

When $t = \pi/2$, $-1 = -\frac{a}{3} - \frac{1}{5}$ or $\frac{a}{3} = \frac{4}{5}$ $\left[\because x\left(\frac{\pi}{2}\right) = -1 \right]$

Hence the solution is $x = \frac{1}{5} (\cos 2t + 4 \sin 3t + 4 \cos 3t)$.

Obs. Laplace transform method can also be used for solving ordinary differential equations with variable coefficients of the form $t^m y^{(n)}(t)$ because $L[t^m y^{(n)}(t)] = (-1)^m \frac{d^m}{ds^m} [L y^{(n)}(t)]$.

Example 21.32. Solve $ty'' + 2y' + ty = \cos t$ given that $y(0) = 1$. (S.V.T.U., 2009)

Solution. Taking Laplace transform of both sides of the equation and noting that

$$L[t f(t)] = -\frac{d}{ds} [L f(t)], \text{ we get}$$

$$-\frac{d}{ds} [s^2 \bar{y} - sy(0) - y'(0)] + 2[s \bar{y} - y(0)] - \frac{d}{ds} (\bar{y}) = \frac{s}{s^2 + 1}$$

or
$$-\left(s^2 \frac{d\bar{y}}{ds} + 2s\bar{y} \right) + y(0) + 0 + 2s\bar{y} - 2y(0) - \frac{d}{ds} (\bar{y}) = \frac{s}{s^2 + 1}$$

or
$$(s^2 + 1) \frac{d\bar{y}}{ds} + 1 = -\frac{s}{s^2 + 1} \quad \text{or} \quad \frac{d\bar{y}}{ds} = -\frac{1}{s^2 + 1} - \frac{s}{(s^2 + 1)^2}$$

On inversion and noting that $L^{-1}\{\bar{f}'(s)\} = -t f(t)$, we get

$$-ty = -\sin t - \frac{1}{2}t \sin t \quad [\text{See § 21.12 (11)}]$$

or
$$y = \frac{1}{2} \left(1 + \frac{2}{t} \right) \sin t$$

which is the desired solution.

Example 21.33. Solve $x \frac{d^2 y}{dx^2} + \frac{dy}{dx} + xy = 0$, $y(0) = 2$, $y'(0) = 0$.

Solution. Taking Laplace transform of both sides of the equation, we get

$$L(xy'') + L(y') + L(xy) = 0$$

or
$$-\frac{d}{ds}[s^2 \bar{y} - sy(0) - y'(0)] + [s \bar{y} - y(0)] - \frac{d\bar{y}}{ds} = 0 \quad \text{or} \quad (s^2 + 1) \frac{d\bar{y}}{ds} + s \bar{y} = 0$$

Separating the variables, $\int \frac{d\bar{y}}{\bar{y}} + \int \frac{s ds}{s^2 + 1} = c$

or
$$\log \bar{y} + \frac{1}{2} \log(s^2 + 1) = \log c' \quad \text{or} \quad \bar{y} = \frac{c'}{\sqrt{s^2 + 1}}$$

Inversion gives $y = c' J_0(x)$

To find c' , we have $y(0) = c' J_0(0)$, i.e., $c' = 2$

Hence $y = 2J_0(x)$.

Example 21.34. An alternating e.m.f. $E \sin \omega t$ is applied to an inductance L and a capacitance C in series.

Show by transform method, that the current in the circuit is $\frac{E\omega}{(p^2 - \omega^2)L} (\cos \omega t - \cos pt)$, where $p^2 = 1/LC$.

Solution. If i be a current and q the charge at time t in the circuit, then its differential equation is

$$L \frac{di}{dt} + \frac{q}{C} = E \sin \omega t \quad [\because R = 0]$$

Taking Laplace transform of both sides, we get

$$L[s\bar{i}(s) - i(0)] + \frac{1}{C}L(q) = E \cdot \frac{\omega}{s^2 + \omega^2}$$

Since $i = 0$ and $q = 0$ at $t = 0$

$$\therefore L[s\bar{i}(s) + \frac{1}{C}L(q)] = \frac{E\omega}{s^2 + \omega^2} \quad \dots(i)$$

Also taking Laplace transform of $i = dq/dt$, we get

$$\bar{i}(s) = L(dq/dt) = sL(q) - q(0)$$

i.e. $L(q) = \bar{i}(s)/s \quad [\because q(0) = 0]$

$$\therefore (i) \text{ becomes } L[s\bar{i}(s) + \frac{1}{C}[\bar{i}(s)/s]] = \frac{E\omega}{s^2 + \omega^2}$$

or
$$\left(Ls + \frac{1}{Cs}\right) \bar{i}(s) = \frac{E\omega}{s + \omega^2} \quad \text{or} \quad \bar{i}(s) = \frac{E\omega s}{L(s^2 + 1/LC)(s^2 + \omega^2)}$$

or
$$\bar{i}(s) = \frac{E\omega}{L} \cdot \frac{s}{(s^2 + p^2)(s^2 + \omega^2)} \quad \text{where } p^2 = 1/LC$$

$$\bar{i}(s) = \frac{E\omega}{L(p^2 - \omega^2)} \left\{ \frac{s}{s^2 + \omega^2} - \frac{s}{s^2 + p^2} \right\}$$

Now taking inverse Laplace transform of both sides, we get

$$i(t) = \frac{E\omega}{L(p^2 - \omega^2)} L^{-1} \left\{ \frac{s}{s^2 + \omega^2} - \frac{s}{(s^2 + p^2)} \right\}$$

or

$$i(t) = \frac{E\omega}{L(p^2 - \omega^2)} (\cos \omega t - \cos pt).$$

PROBLEMS 21.6

Solve the following equations by the transform method :

1. $y'' + 4y' + 3y = e^{-t}$, $y(0) = y'(0) = 1$. (V.T.U., 2008 S ; Kurukshetra, 2005)
2. $(D^2 - 1)x = a \cosh t$, $x(0) = x'(0) = 0$.
3. $y'' + y = t$, $y(0) = 1$, $y'(0) = 0$. (Mumbai, 2009)
4. $y'' - 3y' + 2y = e^{3t}$, when $y(0) = 1$ and $y'(0) = 0$. (V.T.U., 2010)
5. $(D^2 - 3D + 2)y = 4e^{2t}$ with $y(0) = -3$, $y'(0) = 5$. (Mumbai, 2008)
6. $y'' + 25y = 10 \cos 5t$ given that $y(0) = 2$, $y'(0) = 0$. (S.V.T.U., 2008)
7. $(D^2 + \omega^2)y = \cos \omega t$, $t > 0$, given that $y = 0$ and $Dy = 0$ at $t = 0$.
8. $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = \sin t$, $y = \frac{dy}{dt} = 0$ when $t = 0$. (Kurukshetra, 2005 ; Madras, 2003)
9. $\frac{d^4y}{dt^4} - k^4y = 0$, where $y(0) = 1$, $y'(0) = y''(0) = y'''(0) = 0$.
10. $y''''(t) + 2y''(t) + y(t) = \sin t$, when $y(0) = y'(0) = y''(0) = y'''(0) = 0$.
11. $\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 5y = e^{-t} \sin t$, where $y(0) = 0$ and $y'(0) = 1$. (P.T.U., 2010)
12. $y'' + 2y' + 5y = 5y = 5(t - 2)$, $y(0) = 0$, $y'(0) = 0$. (P.T.U., 2005 S)
13. $\frac{d^3y}{dt^3} - 3\frac{d^2y}{dt^2} + 3\frac{dy}{dt} - y = t^2 e^{2t}$, where $y = 1$, $\frac{dy}{dt} = 0$, $\frac{d^2y}{dt^2} = -2$ at $t = 0$.
14. $(D^2 + 1)x = t \cos 2t$, $x = Dx = 0$ at $t = 0$. (Raipur, 2005 ; U.P.T.U., 2005)
15. $ty'' + 2y' + ty = \sin t$, when $y(0) = 1$.
16. $ty'' + (1 - 2t)y' - 2y = 0$, when $y(0) = 1$, $y'(0) = 2$. (P.T.U., 2002)
17. $y'' + 2ty' - y = t$, when $y(0) = 0$, $y'(0) = 1$. (U.P.T.U., 2003)
18. $ty'' + y' + 4ty = 0$ when $y(0) = 3$, $y'(0) = 0$.
19. A voltage Ee^{-at} is applied at $t = 0$ to a circuit of inductance L and resistance R . Show (by the transform method) that the current at time t is $\frac{E}{R - aL} (e^{-at} - e^{-Rt/L})$. (V.T.U., 2000)
20. Workout example 12.17, p. 465 by the transform method.
21. Obtain the equation for the forced oscillation of a mass m attached to the lower end of an elastic spring whose upper end is fixed and whose stiffness is k , when the driving force is $F_0 \sin at$. Solve this equation (using the Laplace transforms) when $a^2 \neq k/m$, given that initial velocity and displacement (from equilibrium position) are zero.

[Hint : The equation of motion is $\frac{d^2x}{dt^2} + \frac{k}{m}x = \frac{F_0}{m} \sin at$ and $x = \frac{dx}{dt} = 0$ when $t = 0$.]

21.16 SIMULTANEOUS LINEAR EQUATIONS WITH CONSTANT COEFFICIENTS

The Laplace transform method can also be applied with advantage to the solution of simultaneous linear differential equations.

Example 21.35. Solve the simultaneous equations $\frac{dx}{dt} + 5x - 2y = t$, $\frac{dy}{dt} + 2x + y = 0$ being given $x = y = 0$ when $t = 0$. [Ex. 13.38]